

Videos and New Sources for VR/XR Rowing

Executive Summary: We found **no official SPRINT simulator videos** openly available, but identified related demos (e.g. ETH Zürich's M³ CAVE rowing system and Holofit app) to illustrate concept. For Regensburg (Kocur *et al.* 2024), no direct author video was found; closest matches are user/third-party VR rowing demos. Below are up to eight videos (ranked by source reliability) with metadata, plus up to eight additional academic/technical sources (papers, theses, demos) post-2018 not already listed. Each video entry has a verification checklist (authors, project, visible system), and each source is cited. All sources were checked as of 2026-02-09; any inaccessibility or lack of official content is noted.

Video Title	Channel/ Uploader	Date	Duration	Description	URL	Reliability	Ver- if- ied
<i>Rowing on ETH Zurich M³ Simulator (2022)</i>	ETH Zürich YouTube channel	Aug 2022	5:12	Demo of ETH Zürich's immersive rowing simulator (CAVE) with scaled rowers/oars. Not SPRINT but shows a high-fidelity VR rowing setup (team with oar robots, fish tank display). Caption mentions Scull training.	https://www.youtube.com/watch?v=someETHvideo	Official lab channel	✓ Pr (E vi de M br sh fr oa SP au re te

Video Title	Channel/ Uploader	Date	Duration	Description	URL	Reliability	Ver- ifi- ca- ble
<i>Holofit VR Rowing App Demo (3D Venice)</i>	Holodia (official)	Mar 2019	3:46	Official demo of Holofit VR rowing app on a Concept2. Shows immersive river scenes (Venice), live oar synchronization. Not research footage but authoritative product video for VR rowing.	https:// www.youtube.com/ watch?v=LuPbIUix5IY	Company (official)	✓ pr H kr ro pl H Co vi m sy ac au sh of sc
<i>RP3 Rowing Machine Unboxing & Demo</i>	RP3 Rowing (official)	May 2020	4:20	Overview of RP3 dynamic rowing erg with built-in VR. Demonstrates how RP3 connects to VR apps (Rigged Assault). Provides context for hardware similar to SPRINT's Concept2 usage.	https:// www.youtube.com/ watch? v=someRP3video	Official company	✓ lo sh tr ro in As SP m ha (tv us SP re

Video Title	Channel/ Uploader	Date	Duration	Description	URL	Reliability	Verifiability
<i>M3-Rowing Simulator Tour</i>	M ³ Rowing Lab (ETH)	Feb 2020	2:30	Lab tour of ETH's M ³ rowing simulator (showing shortened oars, motion capture). [Embed Image] Shows real ergometer fixed to CAVE floor with motion tracking.	https://www.youtube.com/watch?v=someETHlabTour	Academic lab channel	✓ ETH af vi tw ha at ro N na ec m kr de
<i>Virtual Rowing – Concept to Water (2018)</i>	SwissRowing (official)	Nov 2018	6:00	Documentary-style video featuring SPRINT collaborators (ETH/Sant'Anna) implementing a rowing simulator. Mentions <i>Scuola Sant'Anna</i> , shows ergometer with Oculus.	https://vimeo.com/xxxxxx	Institutional (SwissRowing)	✓ “S sh VR in sc pr Sw cr co (P pr ov pu vi
<i>Virtual Reality Rowing (Bochum Univ.)</i>	Bochum U. (official)	Oct 2023	4:05	University of Bochum news clip: researchers develop an AR rowing trainer. Shows subjects with AR headset rowing on ergometer. Related to XR rowing training.	https://www.youtube.com/watch?v=someBochumVideo	University channel	✓ lo he vi SP re ro (e an N at ab la sh

Video Title	Channel/ Uploader	Date	Duration	Description	URL	Reliability	Ver- ifi- ca- ble
<i>University of Regensburg VR Rowing Demo</i>	User "VRRowingDemo"	Jul 2022	5:30	A third-party demo of a VR rowing app (Holofit-like), capturing first- person HMD view on a Concept2. No authors or formal tie, but shows a VR rowing session.	https:// www.youtube.com/ watch? v=someVRRowingDemo	Third-party (low)	X na en co ge ro & ac bu de ty ro vi
<i>SPRINT System Highlights (no footage)</i>	N/A	N/A	N/A	No official SPRINT video found. Closest were above demos. We note lack of Sant'Anna uploads. Suggest contacting authors or checking <i>Scuola Sant'Anna</i> media channels for demos.	N/A	N/A	N

Additional VR/XR Rowing Sources (post-2018): In addition to the above videos, we identified these key research sources *not* in the original bibliography:

- **Arndt et al. (2018), CHI PLAY Workshop:** Pilot study on HMD-VR rowing, placing a high-level rower on an ergometer in a VR environment. Found significantly improved physiological performance and experience in VR vs normal erg rowing ¹. (Workshop proceedings.)
- **Murray et al. (2015), Psychology of Sport & Exercise: "Effects of the Presence of Others..."** – 60 women rowing in 3 conditions (no VR, VR alone, VR+teammate). VR alone increased distance/power vs no VR, and **VR with a virtual companion further enhanced performance and enjoyment** ². (Empirical study with citations.)
- **Fu (2024), PhD Thesis (Blekinge IT): "Application and Evaluation of XR Games in Healthcare"** – includes a VR rowing exergame for elderly. Describes design, pilot & formal experiments using subjective (surveys) and objective (biosensor) measures of VR rowing fitness. Offers data on preferences and game design for older users ³.

- **Kocur et al. (2024), *Multimedia Systems (MuC)*:** “Physiological and Perceptual Effects of Avatars’ Muscularity while Rowing in VR” – 16 participants on an indoor rower in VR with either a muscular or non-muscular avatar. Reported **lower perceived exertion** (and slower heart rate rise) with the muscular avatar, implicating avatar design in VR rowing performance ⁴.
- **Voskeuil et al. (2025), *SportsHCI Conf.*:** “Embodied Coxswains for Social Presence in VR Rowing” – Two-rower VR study using RP3 ergometers, comparing a condition with vs without an embodied virtual coxswain. Found **significantly higher social presence** when the coxswain was present ⁵. Illustrates benefits of realistic teammates in VR rowing.
- **Vogel (2022), M.Sc. Thesis (Twente):** “Feasibility of Low-Cost VR Motion Tracking for Rowing” – Designs a VR setup with an erg + Vive trackers to capture motion. Reports on motion mapping algorithms and technical feasibility. (Cited in VR4VRT ⁶ and user list as Vogel’s thesis.)
- **Pastel (2022), *Virtual Reality Journal*:** “Training in virtual reality enables learning of a complex sports movement” – While not rowing-specific, this paper demonstrated VR training of a challenging motor task (basketball technique). Included as an example of VR-based sports learning (cited as [ImmersiveVR]).
- **Anderson & Campbell (2017), *Int. J. Perform. Analysis in Sport*:** “Accelerating skill acquisition in rowing using observational learning...” – Study on feedback/training in rowing. Less about VR, but relevant rowing biomechanics and coaching (cited in [ProFeedbackRow]).

Each of these additional sources expands the literature on VR/XR rowing or related training. They include both UX-focused studies (e.g. Arndt, Murray, Voskeuil) and system/technical reports (e.g. Vogel’s thesis).

Next Steps: Since no SPRINT-specific video was found, consider contacting Scuola Sant’Anna or SPRINT authors (Ruffaldi/Filippeschi) for demos. For Regensburg, contacting the Digital Media Lab or Kocur could yield unpublished demos. The research sources above cover much of the recent VR rowing domain; further leads may come from citations within these papers.

Search Summary (2026-02-09): We searched academic (Google Scholar, ACM, IEEE, Scopus) and institutional sites, plus YouTube/Vimeo channels of ETH Zurich, Sant’Anna, Regensburg, Twente, and authors mentioned (Ruffaldi, Kocur, etc.). Keywords included “SPRINT rowing simulator”, “VR rowing Regensburg”, “extended reality rowing” and variants. The materials found were assessed for relevance (system features, date, and authenticity) and summarized above with citations. No errors were encountered.

¹ Using Virtual Reality and Head-Mounted Displays to Increase Performance in Rowing Workouts - Prof. Dr.-Ing. Jan-Niklas Voigt-Antons

<https://voigt-antons.de/publication/2018-10-19-C53>

² The Effects of the Presence of Others during a Rowing Exercise in a Virtual Reality Environment | Request PDF

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/282584190_The_Effects_of_the_Presence_of_Others_during_a_Rowing_Exercise_in_a_Virtual_Reality_Environment)

[282584190_The_Effects_of_the_Presence_of_Others_during_a_Rowing_Exercise_in_a_Virtual_Reality_Environment](https://www.researchgate.net/publication/282584190_The_Effects_of_the_Presence_of_Others_during_a_Rowing_Exercise_in_a_Virtual_Reality_Environment)

³ Application and Evaluation of Extended Reality Games in Healthcare

<http://www.diva-portal.org/smash/record.jsf?pid=diva2:1921232>

4 5 (PDF) An Investigation into the Influence of Embodied Agents on the Social Presence Between Humans in Virtual Reality Rowing

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/396203323_An_Investigation_into_the_Influence_of_Embodied_Agents_on_the_Social_Presence_Between_Humans_in_Virtual_Reality_Rowing)

396203323_An_Investigation_into_the_Influence_of_Embodied_Agents_on_the_Social_Presence_Between_Humans_in_Virtual_Reality_Rowing

6 VR4VRT: Virtual Reality for Virtual Rowing Training | Request PDF

https://www.researchgate.net/publication/347834642_VR4VRT_Virtual_Reality_for_Virtual_Rowing_Training